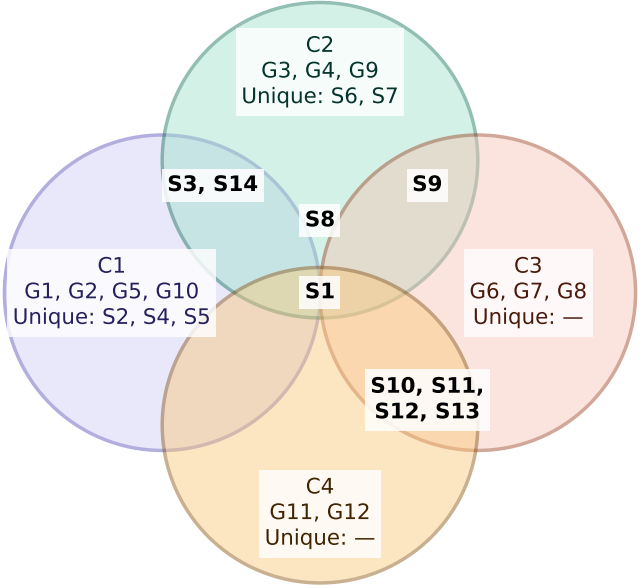

Research gaps

C1 Data Infrastructure

- **G1** Absence of public benchmarks
- **G2** Generalizability & domain shift
- **G5** Annotation scarcity
- **G10** Privacy-preserving & multi-institutional learning

C2 Modeling & Learning

- **G3** Temporal modeling & video-level analysis
- **G4** 2D constraints & 3D reconstruction
- **G9** Class imbalance & rare pathology coverage

C3 Clinical Deployment

- **G6** Clinical validation
- **G7** Explainability & model transparency
- **G8** Multi-modal data integration

C4 Outcome & Reporting

- **G11** Clinical outcome prediction
- **G12** LLMs & structured reporting

Mitigation strategies

- **S1** Large-scale multi-institutional annotated angiographic datasets
- **S2** Dedicated challenge benchmarks with standardized evaluation metrics
- **S3** Self-supervised / foundation model pretraining on unlabeled fluoroscopic video
- **S4** Federated learning with data-use agreements & audit mechanisms
- **S5** Domain adaptation for contrast, dose, frame rate, and motion
- **S6** Video Transformer architectures with long-range attention & SSMs
- **S7** Physics-informed networks with projection geometry & multi-view fusion
- **S8** Semi-supervised, active & self- supervised learning for annotation
- **S9** Uncertainty quantification & OOD failure detection
- **S10** Prospective multi-center RCTs with clinical outcome endpoints
- **S11** Outcome-supervised training for clinically meaningful prediction
- **S12** Longitudinal cohort datasets linking imaging to clinical endpoints
- **S13** Vision-language models for image & free-text report integration
- **S14** Generative augmentation & rare-event simulation pipelines